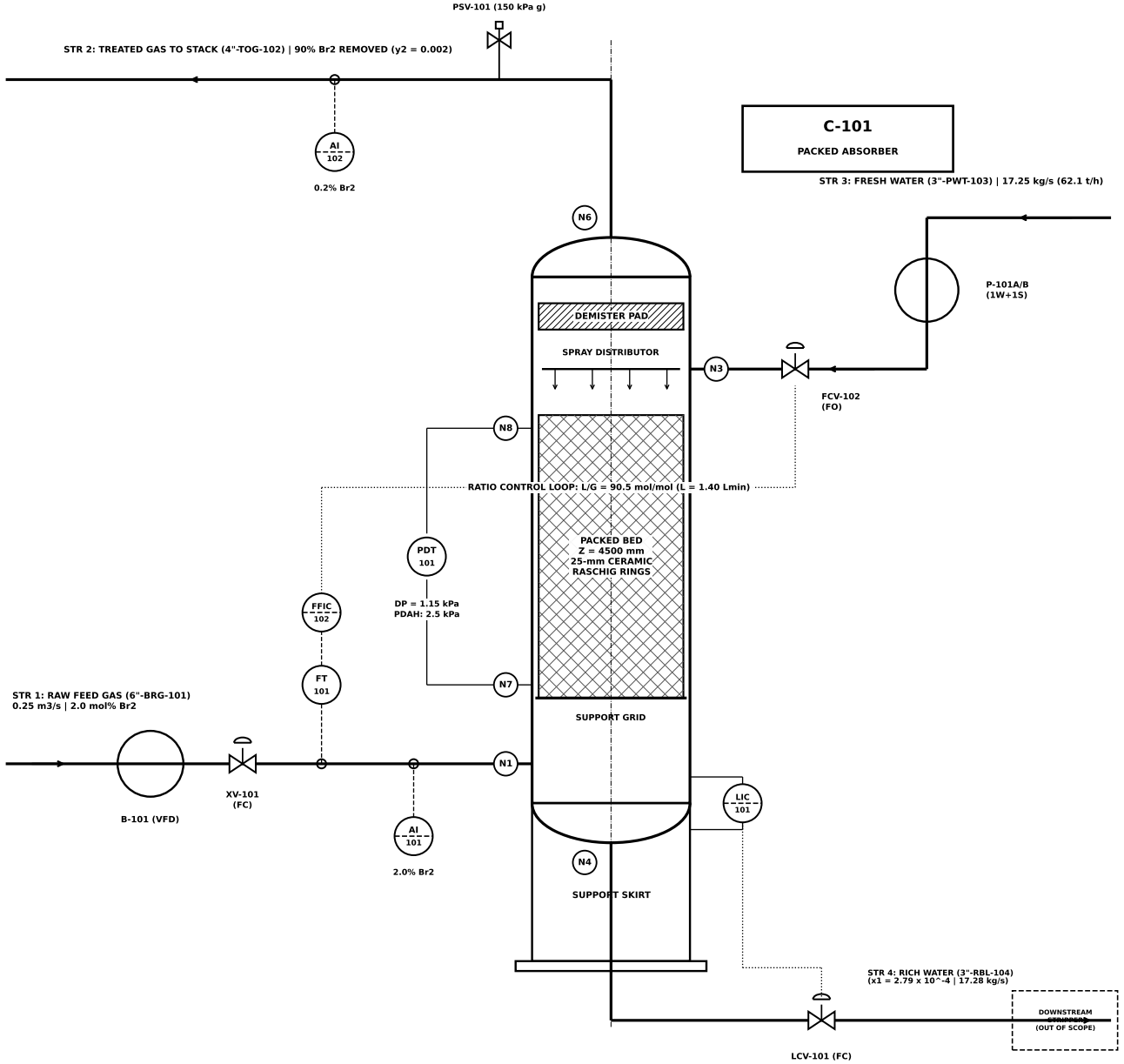


**TABLE 1 — STREAM OPERATING BALANCE**

| STR | SERVICE | MASS FLOW | TEMP | PRESS | COMPOSITION |
|---|---|---|---|---|---|
| 1 | Raw Feed Gas | 0.341 kg/s | 25 DEG C | 803 mmHg | 2.0 mol% Br2 |
| 2 | Treated Vent | 0.315 kg/s | 25 DEG C | 794 mmHg | 0.2 mol% Br2 |
| 3 | Scrub Water In | 17.25 kg/s | 25 DEG C | 2.5 bar g | 100 wt% Water |
| 4 | Rich Liquor Out | 17.28 kg/s | 25 DEG C | 1.1 bar g | x1 = 2.79 x 10^-4 |

**TABLE 2 — PROCESS PIPING LINE SCHEDULE**

| LINE TAG | SERVICE | SIZE | MATERIAL |
|---|---|---|---|
| 6"-BRG-101-CS/PVDF | Raw Bromine Gas Inlet | DN 150 | SA-106 + PVDF |
| 4"-TOG-102-CS/PVDF | Treated Overhead Gas | DN 100 | SA-106 + PVDF |
| 3"-PWT-103-CS | Fresh Scrubbing Water | DN 80 | Carbon Steel |
| 3"-RBL-104-CS/PVDF | Rich Bromine Water Bottoms | DN 80 | SA-106 + PVDF |
| 1.5"-DMW-106-SS | Demister Wash Flush Line | DN 40 | SS316L |

**PROCESS CONTROL PHILOSOPHY & SIS INTERLOCK LOGIC**

**1. RATIO CONTROL LOOP (FFIC-102):**

WATER FLOW (FCV-102) AUTOMATICALLY SLAVED TO RAW GAS INLET (FT-101).

MAINTAINS CONSTANT L/G = 90.5 mol/mol (L = 1.40 Lmin) TO GUARANTEE 90% Br2 RECOVERY.

**2. BED DIFFERENTIAL PRESSURE SAFETY PROTECTION (PDT-101):**

NORMAL OPERATING DP = 1.15 kPa across 4.5 m BED; HIGH ALARM PDAH AT 2.5 kPa (70% FLOOD).

HIGH-HIGH INTERLOCK TRIP AT 3.8 kPa CLOSES XV-101 & TRIPS BLOWER B-101.

**3. BOTTOMS SUMP LEVEL CONTROL (LIC-101):**

MODULATES LCV-101 TO MAINTAIN 50% SUMP LIQUID LEVEL (30 s LIQUID RESIDENCE TIME).

**4. ENVIRONMENTAL & EMISSION MONITORING (AI-102):**

CONTINUOUS Br2 STACK ANALYZER (AI-102) MONITORS OUTLET CONCENTRATION;

ALARMS IF OUTLET Br2 EXCEEDS PROJECT-SPECIFIED DISCHARGE LIMIT.

**5. OVERPRESSURE SAFETY RELIEF (PSV-101):**

SET AT 150 kPa g DISCHARGING TO CAUSTIC EMERGENCY SCRUBBING HEADER SYSTEM.

**JADAVPUR UNIVERSITY | CHEMICAL ENGINEERING DEPARTMENT | PROCESS EQUIPMENT DESIGN & DRAWING**

| DRAWING TITLE | LAB / SUBJECT | COURSE / BRANCH | SCALE |
|---|---|---|---|
| PACKED-BED ABSORBER (C-101) — PIPING & INSTRUMENTATION (P&ID) CAD | PROCESS EQUIPMENT DESIGN & DRAWING | B.E. CHEMICAL ENGINEERING | ISA-5.1 STANDARD |
| **STUDENT NAME** | **ROLL NO.** | **DRAWING NO.** | **DATE** |
| SOURISH SENAPATI | 002310301035 | PEDD-PBA-05 | 29 / 09 / 2026 |

PACKED ABSORBER (C-101) | 0.25 m3/s GAS (2% Br2) | WATER: 17.25 kg/s | ASME SEC VIII DIV 1 / B16.5